

→ Remainder (%,)

$$10 \% 4 \Rightarrow 2$$

$$99 \% 8 \Rightarrow 3.$$

$$\begin{array}{r} 8 \overline{) 99} \quad (12 \\ \underline{96} \\ 3 \end{array}$$

$a \% b \Rightarrow$ remainder when 'a' is divided by 'b'.

$$\text{Dividend} = \text{divisor} \times \text{divisor} + \text{remainder}$$

$$\Rightarrow \text{Remainder} = \text{Dividend} - \underbrace{\text{divisor} \times Q}_{\substack{\downarrow \\ \text{max}^m \text{ multiple of divisor} \\ \leq \text{dividend}}}$$

$$\begin{aligned} &= 99 - \underbrace{8 \times 12}_{\substack{\downarrow \\ 96 \rightarrow \text{max}^m \text{ multiple of } 8 \leq 99}} \end{aligned}$$

$$\text{Quiz 2} \quad -40 \% 7 = -40 - (\text{max}^m \text{ multiple of } 7 \leq -40)$$

$$= -40 - (-42)$$

$$= -40 + 42 = \boxed{2}.$$

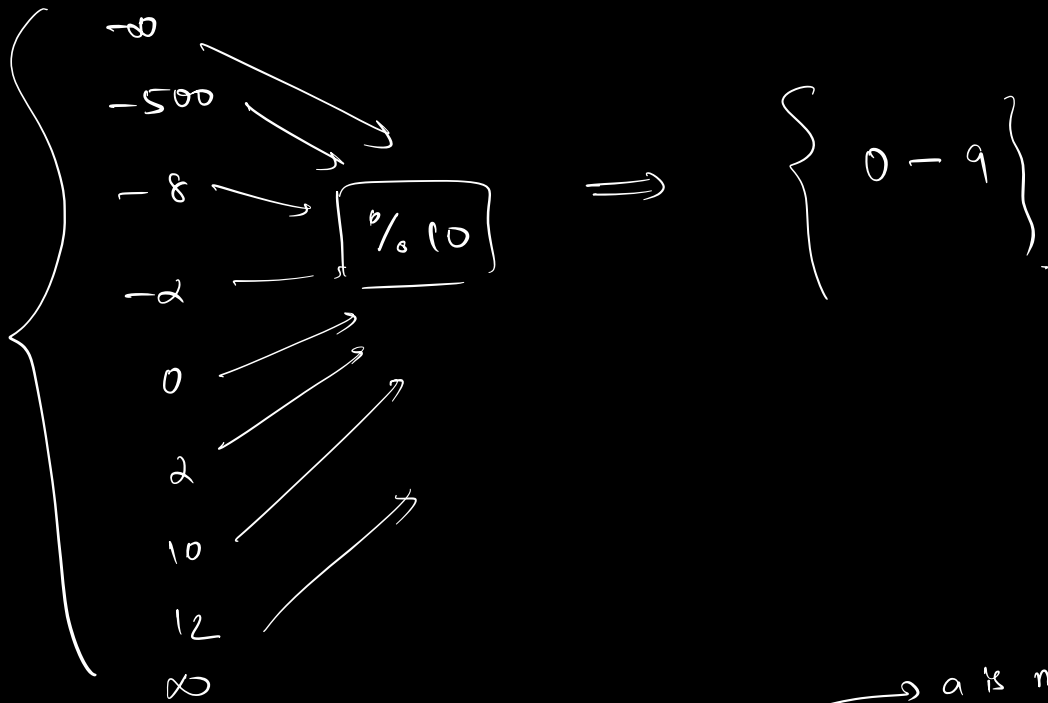
$$\text{Quiz 3} \quad -60 \% 9 = -60 - (\text{max}^m \text{ multiple of } 9 \leq -60)$$

$$= -60 - (-63)$$

$$= \boxed{3}.$$

<u>C/C++ Java</u>	vs	<u>Python</u>
$-401.7 \rightarrow -5 \text{ } 7$	$\longrightarrow$	2
$-607.9 \rightarrow -6 \text{ } 9$	$\longrightarrow$	3

why?



Quiz 4

$$a \rightarrow \% M \Rightarrow$$

$$\left\{ \begin{matrix} 0 \\ \vdots \\ M-1 \end{matrix} \right\}$$

$\rightarrow a$ is multiple of M

Modular Arithmetic

$$1) \quad (a+b) \% M = (a \% M + b \% M) \% M$$

$$11) \quad a \% M = (a+M) \% M \quad \downarrow$$

$$\begin{aligned} & (a \% M + M \% M) \% M \\ &= (a \% M + 0) \% M \\ &= (a \% M) \% M \\ &\quad \downarrow \\ &\quad < M \\ &= a \% M \end{aligned}$$

$$3) \quad (a \times b) \% M = (a \% M \times b \% M) \% M$$

$$4) \quad (a-b) \% M = (a \% M - b \% M + M) \% M$$

Q1. Given $A \geq B$, $A > B$. Find the no. of M such that —
 $A \% M = B \% M$, $M > 1$

ex) $1 \rightarrow A = 16, B = 4$

$M = 2, 3, 4, \cancel{5}, 6,$

$\cancel{7}, \cancel{8}, \cancel{9}, 10, 11, 12,$

$13, 14, 15, 16,$

$$a \equiv b \Rightarrow A=13 \quad B=7, \quad m=2, 3, 6$$

$$\Rightarrow A \% m = B \% m$$

$$\Rightarrow A \% m - B \% m = 0$$

+ add m on both sides

$$A \% m - B \% m + m = m$$

% m on both sides

$$(A \% m - B \% m + m) \% m = m \% m \quad \leftarrow$$

$$\Rightarrow (A - B) \% m = 0$$

possible values of m

for which $(A - B) \% m = 0$

= no. of factors of $(A - B)$

sqrt()

solve

for remaining
1 as factor

(-1)

$$(a - b) \% m = (a \% m - b \% m + m) \% m$$

Q2. Given N array elements, calculate the no. of pairs (i, j)

such that $(arr[i] + arr[j]) \% m = 0$, m is given

Note: $(i \neq j)$, and $((i, j) \text{ \& } (j, i) \text{ is considered as same pair})$

ex 1 $\Rightarrow$ arr[6] $\Rightarrow$ ⁰4 ¹7 ²6 ³5 ⁴5 ⁵3 $m=3$.

pairs	i	j	arr[i]	arr[j]	sum
	0	3	4	5	9 % 3
	0	4	4	5	9 % 3
	1	3	7	5	12 % 3
	1	4	7	5	12 % 3
	2	5	6	3	9 % 3

5

arr $\Rightarrow$ ⁰17 ¹2 ²5 ³4 ⁴6 ⁵23 ⁶12 ⁷26 ⁸14 ⁹18 ¹⁰15 ¹¹30 ¹²35
 $m=10$

i	j	arr[i]	arr[j]	sum
0	5	17	23	40 % 10
0	6	17	12	30 % 10
1	9	2	18	20 % 10
2	10	5	15	20 % 10
2	12	5	35	40 % 10
3	4	4	6	10 % 10
3	7	4	26	30 % 10
4	8	6	14	20 % 10
7	8	26	14	40 % 10
10	12	15	35	50 % 10

opp = 10

Ques 5 $\Rightarrow$ arr[] $\Rightarrow$ [13, 14, 22, 3, 32, 19, 16] $M = 4$

Brute force

check all pairs

```

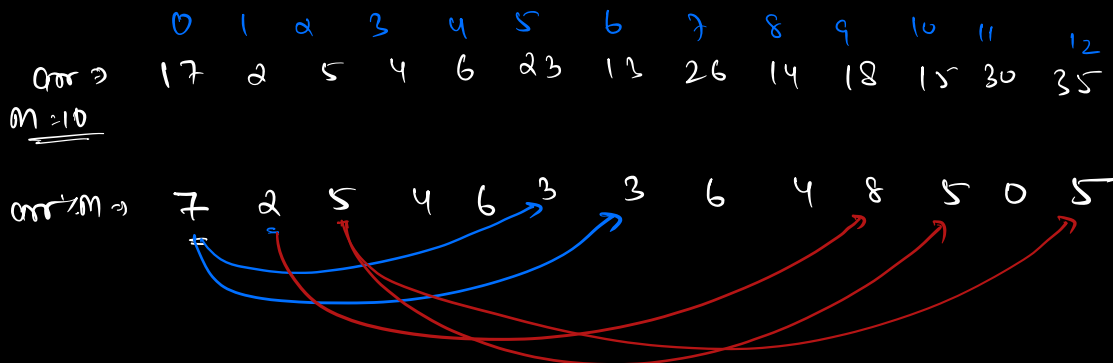
count = 0
for (i = 0; i < N; i++) {
    for (j = i + 1; j < N; j++) {
        if ((arr[i] + arr[j]) % M == 0) {
            count++
        }
    }
}
return count

```

TC $\Rightarrow O(N^2)$ SC $\Rightarrow O(1)$

Obs

$(arr[i] + arr[j]) \% M == 0$
 $\hookrightarrow (arr[i] \% M + arr[j] \% M) \% M == 0$



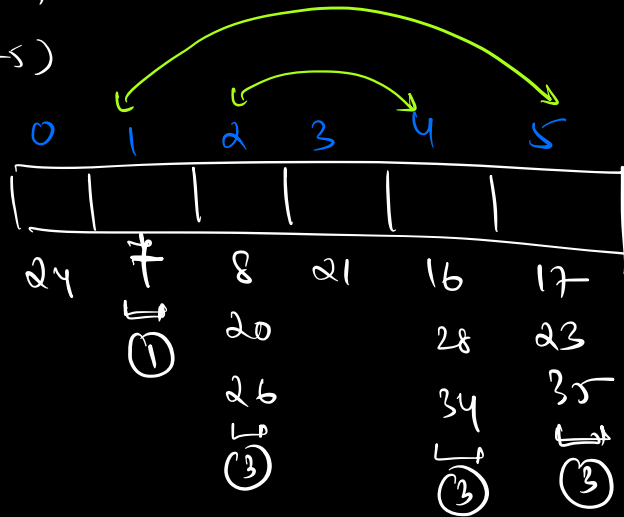
$$(a+b) \% M = (a \% M + b \% M) \% M$$

24

obs

arr[12] $\Rightarrow \{24, 16, 8, 7, 17, 23, 35, 20, 26, 28, 34, 21\}$

$m=6$ (0-5)



arr[1] $\Rightarrow$



$m=6$

remainder $\Rightarrow$



no. of elements

4 2 3 5 2 3

- (4) $\downarrow$
- 6
 - 18
 - 24
 - 30

- 3
- 9
- 15
- 21
- 27

$${}^N C_2 = \frac{N(N-1)}{2}$$

${}^N C_2$

$$\frac{4(4-1)}{2} = 6$$

total $\Rightarrow$

$$(0) \quad (1,5) \quad (2,4) \quad (3)$$

$$4C_2 + 2 \times 3 + 3 \times 2 + 5C_2$$

$$= 6 + 6 + 6 + 10 = 28 \text{ pairs}$$

$$\frac{arr[i] \% m + arr[j] \% m}{2} \% m$$

$6 \% m = 0$

Step 1 - remainder freq

```

hashmap<int, int> freqHM;
for (i=0; i<N; i++) {
    freqHM[arr[i] % m]++;
}

```

} O(N)

Step 2 - count for remainder = 0.

```

ans = hm[0]

```

} O(1)

Step 3

```

i=1, j=M-1
while (i<j) {
    ans = ans + [hm[i] * hm[j]]
    i++
} j--

```

} O(M)

Step 4 $\text{if } (m \% 2 == 0) \{$
 $\text{ans} = \text{ans} + \frac{\text{sum}[m/2]}{2} \}$ $O(1)$
 $\}$

TC $\Rightarrow O(N+M)$
 SC $\Rightarrow O(M)$

Q2. Given an array of all distinct integers, where
 $\{0 \leq \text{arr}[i] \leq N-1\}$, N is size of array.

$\Rightarrow$ Replace $\text{arr}[i]$ with $\text{arr}[\text{arr}[i]]$.

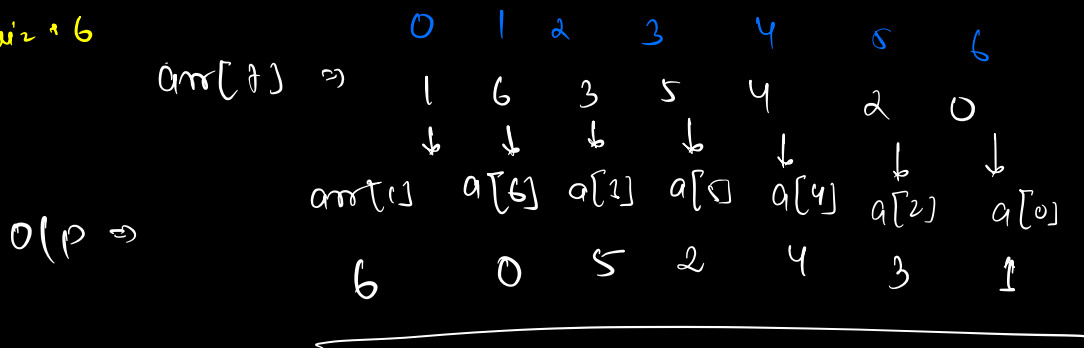
ex $\Rightarrow$ $\text{arr}[5] \Rightarrow$

	0	1	2	3	4
	3	2	4	1	0
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
	$\text{arr}[\text{arr}[0]]$	$\text{arr}[\text{arr}[1]]$	$\text{arr}[\text{arr}[2]]$	$\text{arr}[\text{arr}[3]]$	$\text{arr}[\text{arr}[4]]$
	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$	$\downarrow$
	$\text{arr}[3]$	$\text{arr}[2]$	$\text{arr}[4]$	$\text{arr}[1]$	$\text{arr}[0]$
final $\Rightarrow$	1	4	0	2	3

$\downarrow$
 always update with old value

o/p $\Rightarrow [1, 4, 0, 2, 3]$

Quiz 6



brute force

```

newArr[N]

for (i=0; i<N; i++) {
    newArr[i] = arr[arr[i]]
}

```

$TC \Rightarrow O(N)$
 $SC \Rightarrow O(N)$

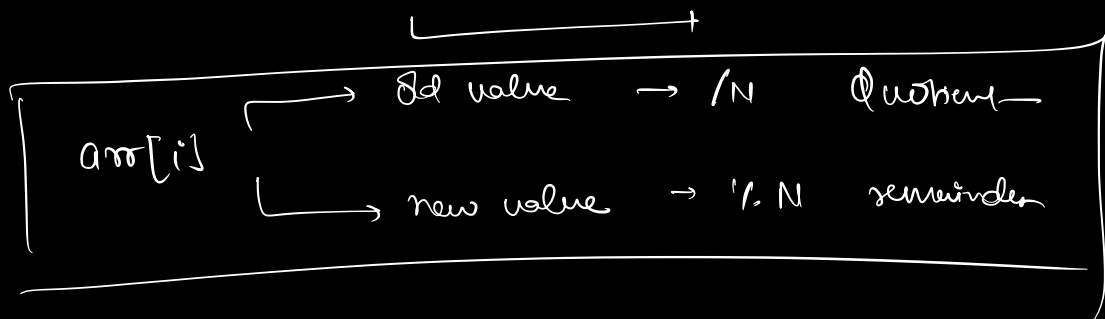
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Day	Hour	
0	0	→ 23 hrs
		46 hrs
		100 hrs
		125 hrs
		x hrs

D	T
0	23
1	22
4	4
5	5
$x/24$	$x\%24$

$arr[i]$
 ↓
 $(arr[arr[i]])$
 ↓
 idx
 ↘
 val
 =

total hours $\Rightarrow 24 \times D + T$



arr[i] $\Rightarrow$

0	1	2	3	4
3	2	4	1	0

multiply * 5

$3 \times 5 = 15$	$2 \times 5 = 10$	$4 \times 5 = 20$	$1 \times 5 = 5$	$0 \times 5 = 0$
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idx [arr[i]]

$15/5 = 3$	$10/5 = 2$	$20/5 = 4$	$5/5 = 1$	$0/5 = 0$
------------	------------	------------	-----------	-----------

val arr[idx] % 5

1	4	0	2	3
---	---	---	---	---

$\downarrow$
[arr[i]]

new arr $\Rightarrow$

0	1	2	3	4
$3 \times 5 = 15$	$2 \times 5 = 10$	$4 \times 5 = 20$	$1 \times 5 = 5$	$0 \times 5 = 0$

$\%N \Rightarrow$

1	4	0	2	3
---	---	---	---	---

Step 1 * multiply array with N

step 2 → update the array

for ($i=0$; $i < N$; $i++$) {

$idx = arr[i] / N$

$val \Rightarrow arr[idx] / N$ ← original val. at idx

$arr[i] = arr[i] + val$

}

step 3 → create the final result

for ($i=0$; $i < N$; $i++$) {

$arr[i] = arr[i] \% N$

}

$O(P)$

HW

Given $arr[N]$ $[0 \leq arr[i] \leq N-1]$

store two values → $arr[i]$, freq of $arr[i]$

TC $\Rightarrow O(N^2)$

new array → element_N → original $arr[i]$

SC $\Rightarrow O(1)$

element_{%N} → freq of $arr[i]$

$\Rightarrow$ Inverse modulo:-

$$(a/b) \% M = \frac{a \% M}{b \% M}$$

$\downarrow$

$$(a \times b^{-1}) \% M \Rightarrow [(a \% M) \times (b^{-1} \% M)] \% M.$$

Inverse Modulo $\Rightarrow$ given a, m , b can be an IV of a if,

$$\frac{(a \times b) \% M = 1}{\text{greatest-common divisor}}$$

$\Rightarrow$ IV only exists if $\gcd(a, m) = 1$

$\Rightarrow$ value of b : $[1, m-1]$.

ex $\Rightarrow$ $a = 10, m = 7$ $\boxed{b} \rightarrow$ IV exists or not

$$\gcd(a, m) = 1$$

$$\gcd(10, 7) = 1 \quad \checkmark$$

$$IV(b) \rightarrow [1, 6]$$

$$b=1 \rightarrow (10 \times 1) \% 7 = 3$$

$$b=2 \rightarrow (10 \times 2) \% 7 = 6$$

$$b=3 \rightarrow (10 \times 3) \% 7 = 2$$

$$b=4 \rightarrow (10 \times 4) \% 7 = 5$$

$$b=5 \rightarrow \boxed{(10 \times 5) \% 7 = \textcircled{1}}$$

$$\underline{\underline{IV = 5}}$$

↓

$$\boxed{a^{-1} \% m = 5}$$

← IV

$$a = 10$$

$$m = \underline{\underline{7}}$$

aim find IV.

if (gcd(a, m) == 1) {

for (i = 1; i < m; i++) {

if ((a * i) % m == 1)

return i

} }